

EASTERN IOWA, IA, USA

WMO: 725450

Lat: 41.883N

Lon: 91.717W

Elev: 868

StdP: 14.24

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14990

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.8	(c) -3.3	(d) -14.8	(e) 2.5	(f) -8.1	(g) -9.7	(h) 3.4	(i) -2.6	(j) 32.5	(k) 20.6	(l) 28.5	(m) 16.3	(n) 9.8	(o) 310	(p) 0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	90.1	76.4	87.5	74.7	84.8	73.0	79.0	86.6	77.2	84.4	75.3	82.2	10.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.8	144.2	84.2	74.9	135.2	82.0	73.1	126.8	79.9	43.2	86.6	41.2	84.2	39.4	82.5	88.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 26.9	(b) 24.0	(c) 20.6	(e) -16.8	(f) 94.1	(g) 6.0	(h) 3.3	(i) -21.1	(j) 96.5	(k) -24.6	(l) 98.4	(m) -28.0	(n) 100.3	(o) -32.3	(p) 102.6		
(4) 26.9	(5) 24.0	(6) 20.6	(7) -17.2	(8) 82.1	(9) 5.9	(10) 1.8	(11) -21.4	(12) 83.4	(13) -24.9	(14) 84.5	(15) -28.2	(16) 85.5	(17) -32.4	(18) 86.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.7	19.9	24.1	36.6	49.3	60.4	70.0	73.1	71.1	64.1	51.5	37.8	25.5
	DBStd	21.01	12.92	12.72	11.84	9.63	8.35	6.22	5.56	5.33	8.01	9.69	10.43	11.96
	HDD50	3488	932	727	442	125	13	0	0	0	3	99	386	761
	HDD65	6718	1397	1146	883	477	193	24	5	10	112	431	816	1224
	CDD50	3028	0	1	26	104	334	600	715	653	427	146	20	2
	CDD65	783	0	0	1	6	50	174	255	198	86	13	0	0
	CDH74	6806	0	0	9	98	448	1494	2304	1573	764	115	1	0
	CDH80	2074	0	0	0	19	104	467	781	452	226	25	0	0
	WSAvg	9.9	11.5	11.4	11.6	12.2	10.4	8.6	7.2	6.7	7.9	9.7	10.8	10.6
	PrecAvg	34.4	1.0	1.0	2.2	3.3	3.9	4.6	4.5	4.1	3.7	2.3	2.0	1.6
(15) Precipitation	PrecMax	60.5	2.6	3.3	6.1	5.7	8.7	11.5	17.0	13.1	10.2	5.6	6.5	3.6
	PrecMin	19.3	0.0	0.1	0.2	0.7	0.4	0.2	0.4	0.4	0.0	0.2	0.2	0.2
	PrecStd	9.1	0.7	0.7	1.4	1.3	2.1	2.7	3.4	2.9	2.3	1.4	1.4	0.9
	PrecStd	9.1	0.7	0.7	1.4	1.3	2.1	2.7	3.4	2.9	2.3	1.4	1.4	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.4	59.6	74.9	83.1	87.9	91.3	95.1	92.8	90.1	83.4	70.1	59.0
		MCWB	44.9	50.8	61.2	64.4	71.4	76.0	78.8	76.7	73.6	69.0	57.9	55.6
	2%	DB	44.7	51.9	66.7	76.3	82.7	88.5	90.3	88.1	85.9	76.7	63.2	51.7
		MCWB	40.7	45.8	55.7	60.6	67.4	74.2	78.3	76.2	71.9	63.9	54.9	48.0
	5%	DB	39.4	45.3	60.8	71.0	79.0	85.5	87.3	84.8	82.2	72.0	57.9	45.7
		MCWB	36.4	41.1	53.5	57.6	65.4	72.5	76.3	74.4	69.6	62.3	51.5	42.4
	10%	DB	35.8	40.3	54.2	65.9	75.3	82.2	84.3	81.9	78.4	67.2	53.5	40.5
		MCWB	33.7	37.2	46.9	54.4	63.7	70.7	74.2	72.3	67.3	58.6	48.3	37.5
	0.4%	WB	46.2	53.4	62.7	66.3	73.6	78.9	82.4	80.6	77.0	71.3	60.5	56.7
		MCDB	48.9	58.4	72.2	77.6	83.9	87.9	90.0	88.6	85.1	79.2	66.1	59.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	40.5	46.4	58.8	62.7	70.8	76.8	80.0	78.0	74.3	67.4	56.6	48.2
		MCDB	44.5	51.1	64.5	72.2	79.5	85.4	87.4	85.2	82.1	73.5	61.5	50.4
	5%	WB	36.8	41.4	53.5	59.7	68.3	74.7	77.9	76.2	71.9	63.8	52.8	42.2
		MCDB	39.3	45.2	60.2	69.1	75.9	82.5	84.7	82.6	79.0	68.6	57.4	45.1
	10%	WB	33.8	37.3	47.6	56.0	65.7	72.6	75.9	74.4	69.7	60.1	48.0	37.7
		MCDB	35.4	40.2	53.5	64.3	72.7	79.9	82.6	80.0	75.9	66.5	52.5	40.4
(35) Mean Daily Temperature Range	5% DB	MDBR	15.7	16.4	18.5	21.1	20.0	19.1	19.0	19.3	21.9	20.2	17.8	15.2
		MCDBR	20.7	21.8	26.5	29.1	23.9	22.0	20.7	21.4	24.7	25.1	23.9	20.6
		MCWBR	16.4	16.0	16.6	16.4	12.4	10.7	10.7	11.1	11.9	14.0	16.4	16.3
	5% WB	MCDBR	18.9	21.2	23.7	25.0	20.8	19.5	19.0	19.0	20.5	20.4	21.7	18.9
		MCWBR	15.5	16.3	15.9	15.7	12.6	11.1	11.1	11.1	11.2	14.1	16.6	16.0
(40) Clear-Sky Solar Irradiance	taub		0.286	0.304	0.331	0.377	0.416	0.422	0.436	0.416	0.385	0.338	0.299	0.286
	taud		2.400	2.361	2.390	2.278	2.235	2.268	2.213	2.277	2.326	2.460	2.494	2.427
	Ebn at Noon		275	286	291	283	273	271	266	267	268	271	269	265
	Edh at Noon		25	30	34	41	43	42	44	40	36	28	23	23
	RadAvg		580	898	1202	1489	1759	1947	1958	1715	1443	949	637	469
(45) All-Sky Solar Radiation	RadStd		63	78	113	120	124	182	133	108	112	111	52	48

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page